



Habib University

School of Science & Engineering

Student Name/ID	
Course:	EE - 211 - Basic Electronics
Semester	Spring 2025
Midterm #	1
Date, Time, Location :	March 11, 2025, 1030-1130, H-503
Instructor :	Syed Arsalan Jawed
Total Marks :	100

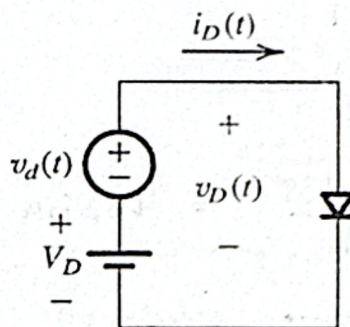
(85/100)
+ 10
85

Instructions for Students

1. Provide your answers within the empty spaces given after each question.
2. Keep your answers concise. Provide neat and tidy calculations, equations and plots.
3. You may consult ONLY your written class lecture notes during the exam.

Question # 1 : [CLO-1]

For the following circuit, start from the given equation and derive the expressions for diode's incremental resistance. [20]



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$$i_D(t) = I_S e^{(V_D + v_d)/V_T}$$

$$v_D(t) = V_D + v_d(t)$$

$$I_0 = I_S e^{V_D/V_T}$$

$$i_D(t) = I_S e^{V_D/V_T} \cdot e^{v_d/V_T}$$

$$i_D(t) = I_0 \cdot e^{v_d/V_T}$$

v_d is a small signal change, for a very small change:

$$e^{v_d/V_T} = \left(1 + \frac{v_d}{V_T}\right)$$

$$i_D(t) = I_0 \cdot \left(1 + \frac{v_d}{V_T}\right)$$

$$i_D(t) = I_0 + I_0 \frac{v_d}{V_T}$$

$$i_D(t) = I_0 + i_d$$

$$\text{hence } i_d = \frac{I_0 V_d}{V_T}$$

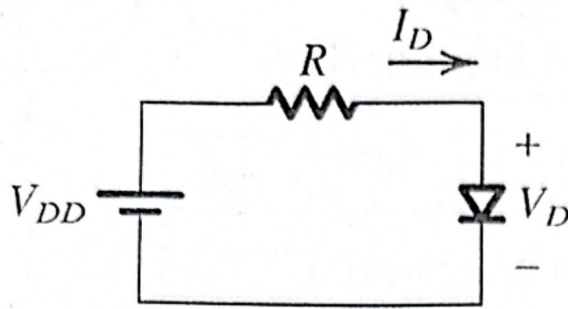
$$\frac{i_d}{v_d} = \frac{I_0}{V_T}$$

$$\frac{1}{r_d} = \frac{I_0}{V_T} \rightarrow \text{Conductance}$$

$$r_d = \frac{V_T}{I_0} \rightarrow \text{resistance}$$

Question # 2 : [CLO-2]

Determine I_D and V_D for the following circuit using iterative analysis. $V_{DD}=10V$, $R=2\text{ k}\Omega$, $V_D=0.7V@1mA$. [20]



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$$I_D = \frac{V_{DD} - V_D}{R} = \frac{10 - 0.7}{2k}$$

$$I_D = \frac{9.3}{2k} = 4.65\text{ mA} \quad \checkmark$$

$V_T \approx 25\text{ mV}$ ✓

$$V_2 - V_1 = 2.3 V_T \log \left(\frac{I_2}{I_1} \right)$$

$$V_2 - 0.7 = 2.3 V_T \log \left(\frac{4.65}{1} \right)$$

$$V_2 = 0.0384 + 0.7$$

$$V_2 = 0.738 \quad \checkmark$$

Second iteration.

$$I_D = \frac{10 - 0.738}{2k} = 4.63\text{ mA}$$

$$0.738 - V = 2.3 V_T \log \left(\frac{4.63}{4.65} \right)$$

$$0.738 - V = -1.07758 \times 10^{-4}$$

$$0.738 + (1.07758 \times 10^{-4}) = V \quad \checkmark$$

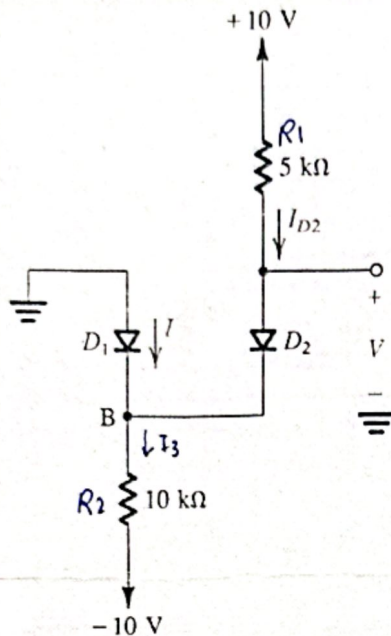
$V = 0.738 \rightarrow$ iteration stops since this value is same as previous iteration.

$$I_D = 4.63\text{ mA}$$

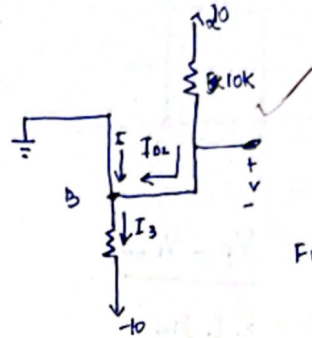
$$V = 0.738\text{ V} \quad \checkmark$$

Question # 3 : [CLO-2]

Find I and V for the following circuit. Assume diodes have a constant voltage drop of 10mV if they are forward biased for any current value. Replace +10V with +20V supply and 5k-ohm resistor with 10k-ohm. A first-order rapid analysis is required. [20]



Assuming both diodes are conducting.



Assuming
V drop = 10mV ✓

Finding I_{D2} :

$$\frac{10 - 0.01}{10k} = \frac{20 - 0.01}{10k} = 1.99 \text{ mA}$$

Applying KCL at node B:

$$I + I_{D2} = I_3$$

$$I + 1.99 \text{ mA} = 0 - (-10)$$

$$I + 1.99 = 1 \text{ mA}$$

$$I = 1 - 1.99$$

$I = -0.99 \text{ mA}$ → since this is not possible, D_1 is not conducting.

Now Assuming only D_2 conducts.

Finding I_{D2} again.

$$\frac{20 - 0.01 - (-10)}{10k + 10k} = \frac{29.99}{20k} = 1.4995 \text{ mA}$$

Finding V :

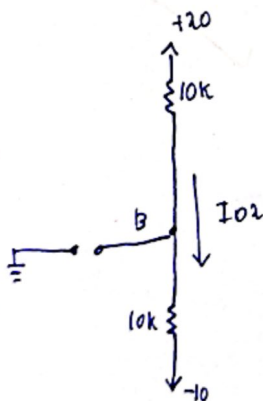
current through 10kΩ resistor = 1.4995 mA

$$V \text{ across } R_2 = V = IR = (10k)(1.4995 \text{ mA}) = 14.995 \text{ V}$$

$$14.995 - 10 = 4.995 \text{ V}$$

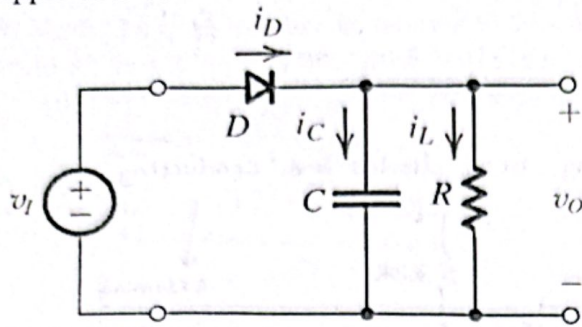
$$I = 1.4995 \text{ mA}$$

$$V = 4.995 \text{ V}$$



Question # 4 : [CLO-3]

Consider a peak rectifier fed by a 60-Hz sinusoid having a peak value $V_p = 100$ V. Let the load resistance $R = 10$ kOhm. Find the value of the capacitance C that will result in a peak-to-peak ripple of 2 V. [20]



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$$V_R = \frac{V_p - V_{D(on)}}{R C f_{in}}$$

$$2 = \frac{100 - 0.7}{(10 \times 10^3) C (60) \times 2\pi}$$

$$C = \frac{100 - 0.7}{10000 \times 60 \times 2} = \frac{99.3}{1200000}$$

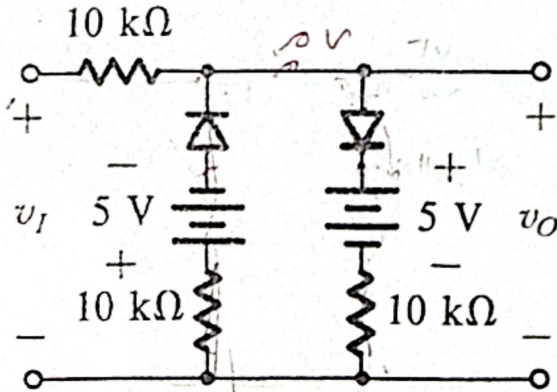
$$= 8.275 \times 10^{-5} \text{ F}$$

$$= 82.75 \mu\text{F}$$

Question # 5 : [CLO-3]

[20]

Plot Voltage Transfer Characteristic plots for the following circuits. Assume diodes are real and have a reverse breakdown voltage of 1V (although practical VBD is higher but for the sake of argument in the question) and a forward bias drop of 2V. For the VTC, sweep V_I from -10V to +10V. Short the 10k-ohm resistors under the 5V dc supplies.



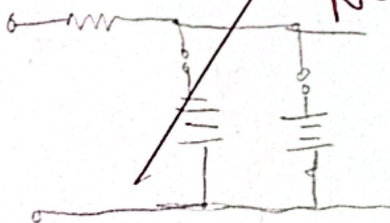
$$\frac{v_{in} - v_o}{10k} + \frac{5 - v_o}{10k} = 0$$

$$v_{in} - v_o - v_o + 5 = 0$$

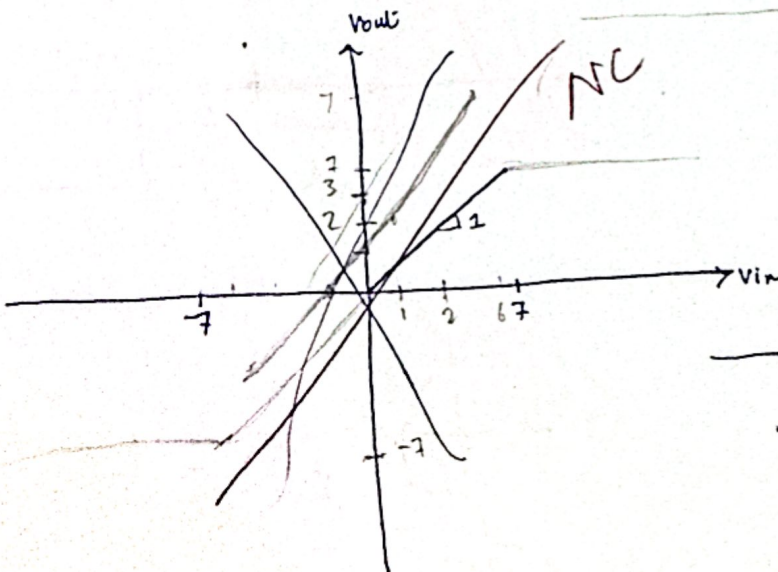
$$v_{in} - 2v_o = -5$$

$$v_o = \frac{5 + v_{in}}{2}$$

NC



$$v_{out} = v_{in} + 1$$



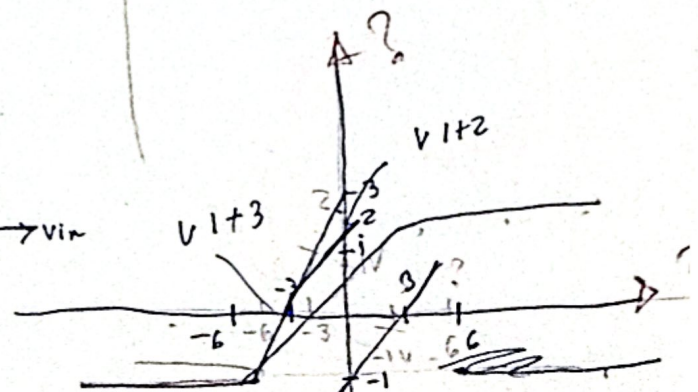
During the positive cycle initially ~~when~~ both diodes are off.

Since the diodes have drop of 2V, the voltage of the anode should be greater than the voltage of cathode by 2V for the diode to conduct.

When v_{in} is between -7 to 7, both diodes are off, so $v_{out} = v_{in}$.

when $v_I > 6V$
-1

when $v_I < -6V$
-1V



Not clear.
Give viva to
get withheld